A random-key metaheuristic for the block relocation problem

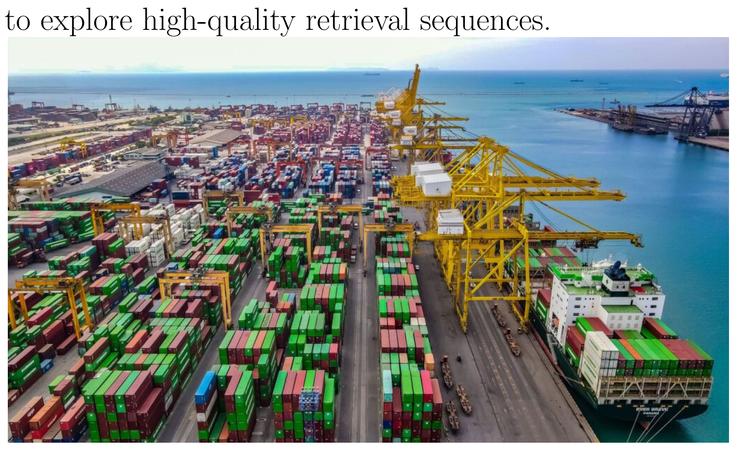
Guilherme Nahass Mateus Martin

Universidade Federal de São Carlos (UFSCar), Campus Sorocaba
gnahass@estudante.ufscar.br mpmartin@ufscar.br

INTRODUCTION

Efficient material handling is a critical problem in **maritime terminals and steel yards**, where **containers, steel coils, and slabs** are stacked due to limited space. Since retrieval must follow strict departure or production schedules, poor stack organization leads to costly reshuffling operations.

We address the **restricted** version of the **unloading block relocation problem** and propose a **decoder within the Random Key Optimizer (RKO)** framework to explore high-quality retrieval sequences.



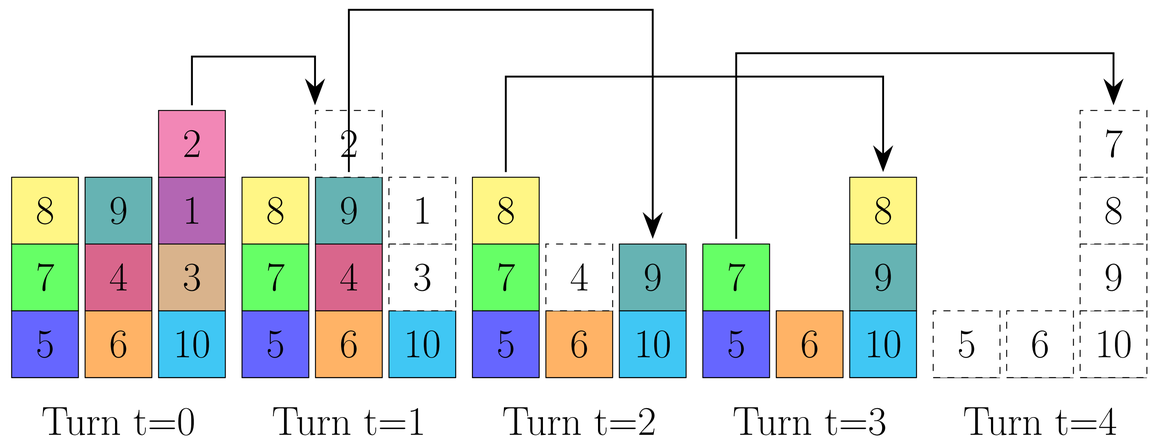
Source: https://www.ectt.co.th/ectt-service/container-yard/.

PROBLEM DESCRIPTION

The restricted unloading block relocation problem is a **combinatorial optimization problem** defined on m stacks with capacity H, containing n identical blocks with unique retrieval priorities $\{1, \ldots, n\}$.

Blocks must be retrieved in strictly increasing priority order. If the next required block is on top of a stack, it is immediately removed. Otherwise, the blocks above it must be relocated to other non-full stacks. In the **restricted** variant, only *forced relocations* are allowed: a block may be moved only if it directly obstructs the current target block. Voluntary (anticipatory) relocations are forbidden.

Objective. Determine a sequence of feasible relocations that retrieves all blocks while minimizing the total number of relocations.



RKO FRAMEWORK & DECODER DESIGN

Random Key Optimizer (RKO) of Chaves et al. (2025)
RKO searches in a continuous space of *random keys* $\mathbf{x} \in [0,1]^n$. Metaheuristics (e.g., BRKGA, ILS, VNS) evolve $\mathbf{x}$, while all problem-specific logic is delegated to a **decoder**.

Decoder Role
The decoder is the mandatory problem-dependent component that:

- Maps random keys to a feasible solution;
- Guarantees feasibility during construction;
- Returns the cost evaluated by the optimizer.

The optimizer never manipulates problem moves directly.

Our Decoder (one key per block)
Each block b is assigned a key $x_b \in [0,1]$. Keys do not encode explicit relocation sequences; instead, they **bias stack selection** during a yard simulation. The RKO thus explores different relocation policies induced by $\mathbf{x}$.

Input: Instance (m, H, n), keys $\mathbf{x} \in [0,1]^n$
Output: Total number of relocations
$relocations \leftarrow 0$
while blocks remain **do**
  $b^* \leftarrow$ current minimum-priority block
  while b^* is blocked **do**
    $b \leftarrow$ top block above b^*
    $S \leftarrow$ non-full stacks (excluding current)
    Rank S by composite score using x_b
    Move b to best stack in S
    $relocations \leftarrow relocations + 1$
  end while
  Dispatch b^*
end while
return $relocations$

PRELIMINARY RESULTS

We tested benchmark instances from the literature (Caserta et al., 2012).

Instance	W	H	N	Time (s)	Best	Avg	Std
data3-8-1-3	8	5	24	10	18	18.00	0.00
data5-5-12-3	5	7	25	10	26	26.00	0.00
data6-6-3-3	6	8	36	10	53	53.00	0.00
data10-6-1-4	6	12	60	10	143	143.00	0.00
data5-5-4-3	5	7	25	10	42	42.00	0.00

Runs: 5 per experiment; Repetitions: 3.
All instances presented zero standard deviation across repetitions, indicating stable decoder behavior.

CONCLUSIONS & NEXT STEPS

Conclusions

- Integration of the **proposed decoder** into the RKO framework.
- Effective mapping of random keys to **relocation policies** using a single key per block.
- Preliminary results show **stable convergence** and high-quality solutions for small-scale instances.

Next steps

- Extensive benchmarking against state-of-the-art algorithms on **large-scale instances**.
- Comparative analysis between **restricted and unrestricted** problem variants.

ACKNOWLEDGMENTS & REFERENCES

Acknowledgments. Supported by FAPESP [grants numbers #2025/11059-3, #2022/05803-3].

Marco Caserta, Silvia Schwarze, and Stefan Voß. A mathematical formulation and complexity considerations for the blocks relocation problem. *European Journal of Operational Research*, 219:96–104, 5 2012. ISSN 03772217. doi: 10.1016/j.ejor.2011.12.039.

Antonio A. Chaves, Mauricio G. C. Resende, Martin J. A. Schuetz, J. Kyle Brubaker, Helmut G. Katzgraber, Edilson F. de Arruda, and Ricardo M. A. Silva. A random-key optimizer for combinatorial optimization. *Journal of Heuristics*, 31(4):32, 2025. ISSN 1572-9397. doi: 10.1007/s10732-025-09568-z.